

**WILSON MOSES**

**DATA SCIENCE × AI ENGINEERING**



**DATA SCIENCE FIELD NOTES / 01**

# EMPLOYEE ATTRITION ANALYSIS

Patterns associated with workforce turnover in 1,470 employee records

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AnalystLab Africa | Data Science Internship | Week 1

Learning. Building. Applying.

## BUSINESS CONTEXT

# Understand the pattern before predicting it

### THE BRIEF

ABC Manufacturing wants to understand where attrition is concentrated before investing in predictive modelling. The analysis examines workforce structure, overtime, departments, job roles, age, income and tenure.

### DATASET

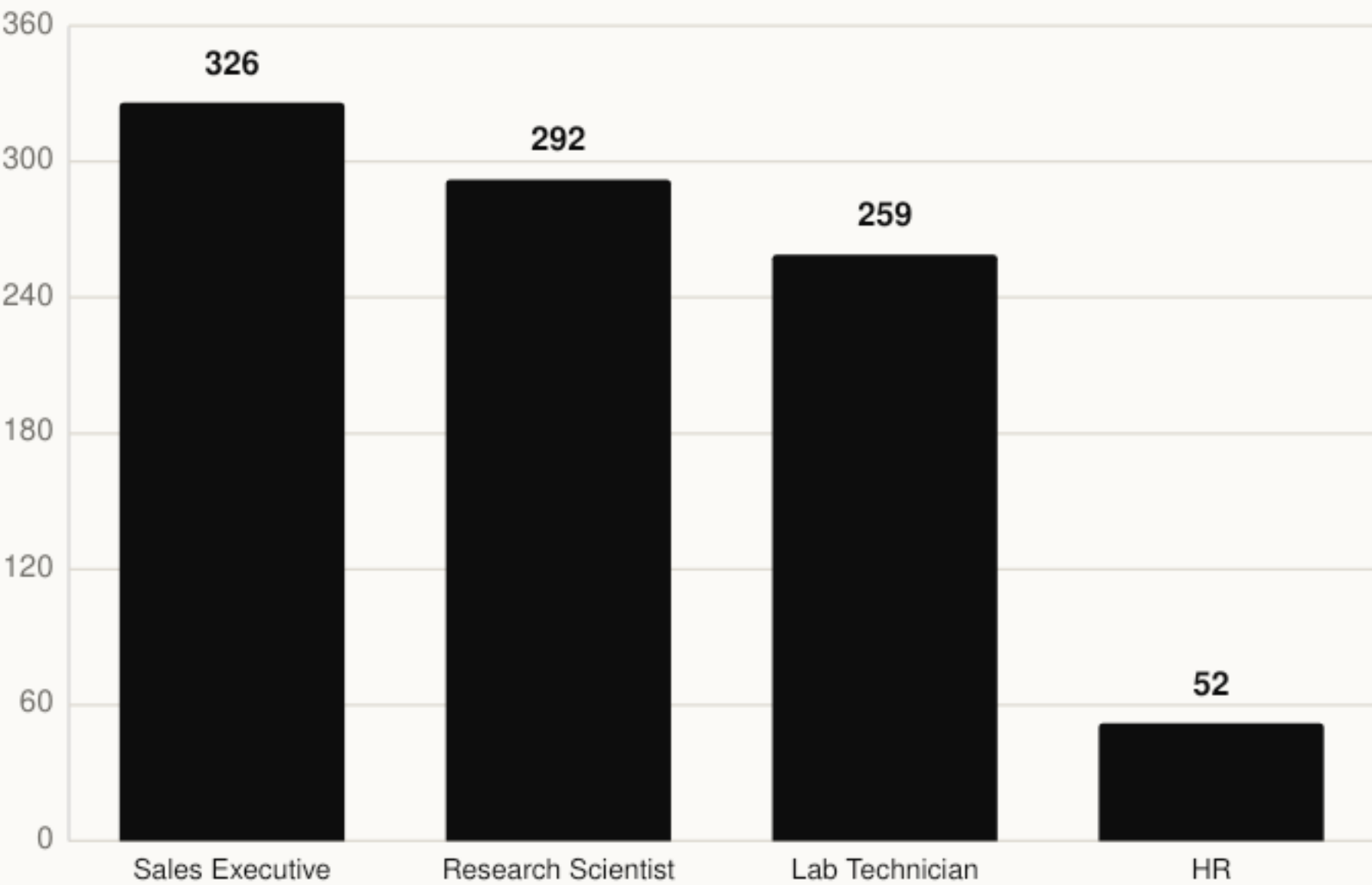
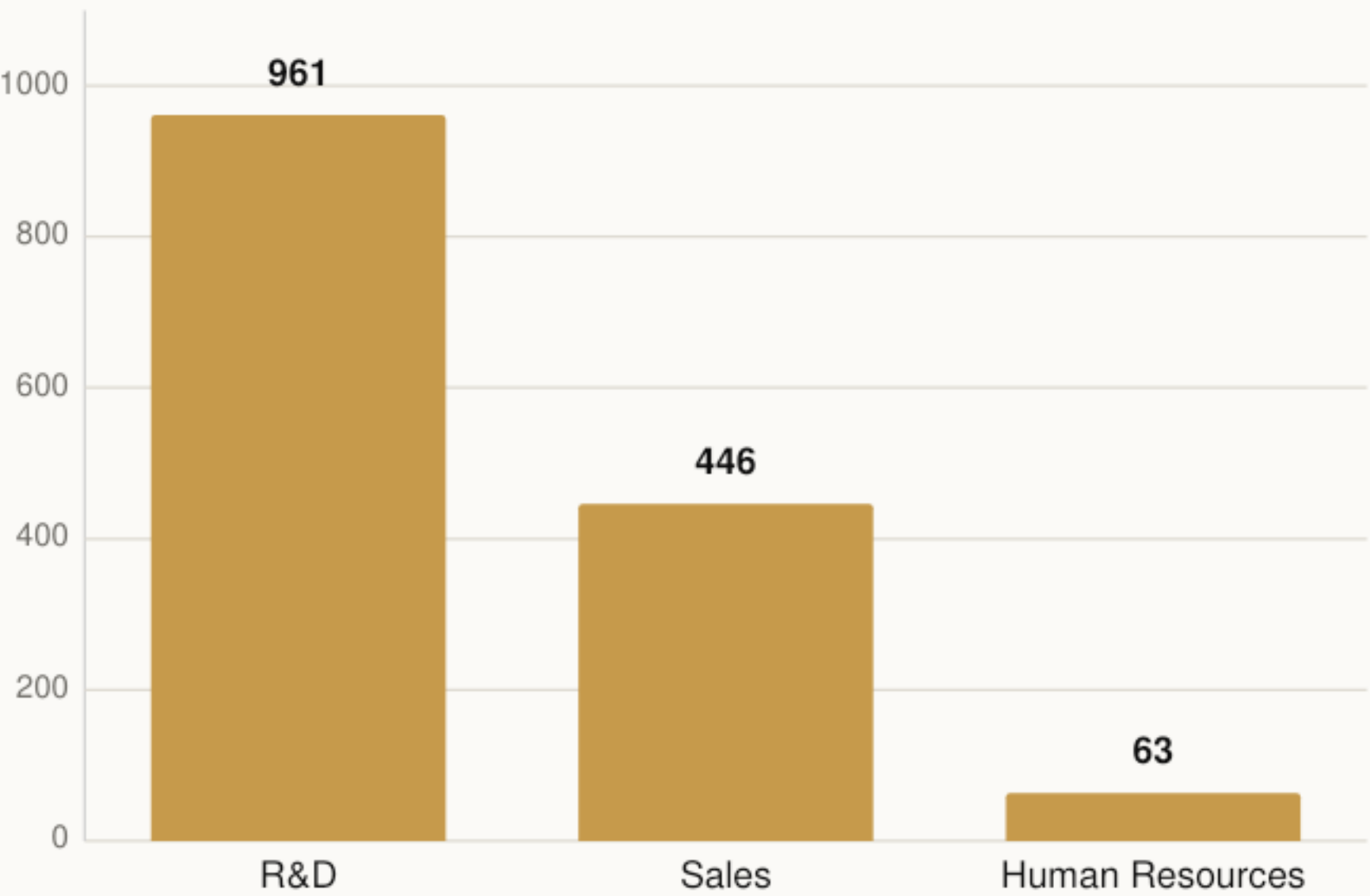
**1,470 employees • 35 variables • 0 missing values • 0 complete duplicates**

### QUESTIONS

- 01** What does the workforce look like?
- 02** Which departments exceed the baseline?
- 03** How strongly is overtime associated with attrition?
- 04** Which roles show the highest turnover?
- 05** How do age, income and tenure differ between groups?

WORKFORCE OVERVIEW

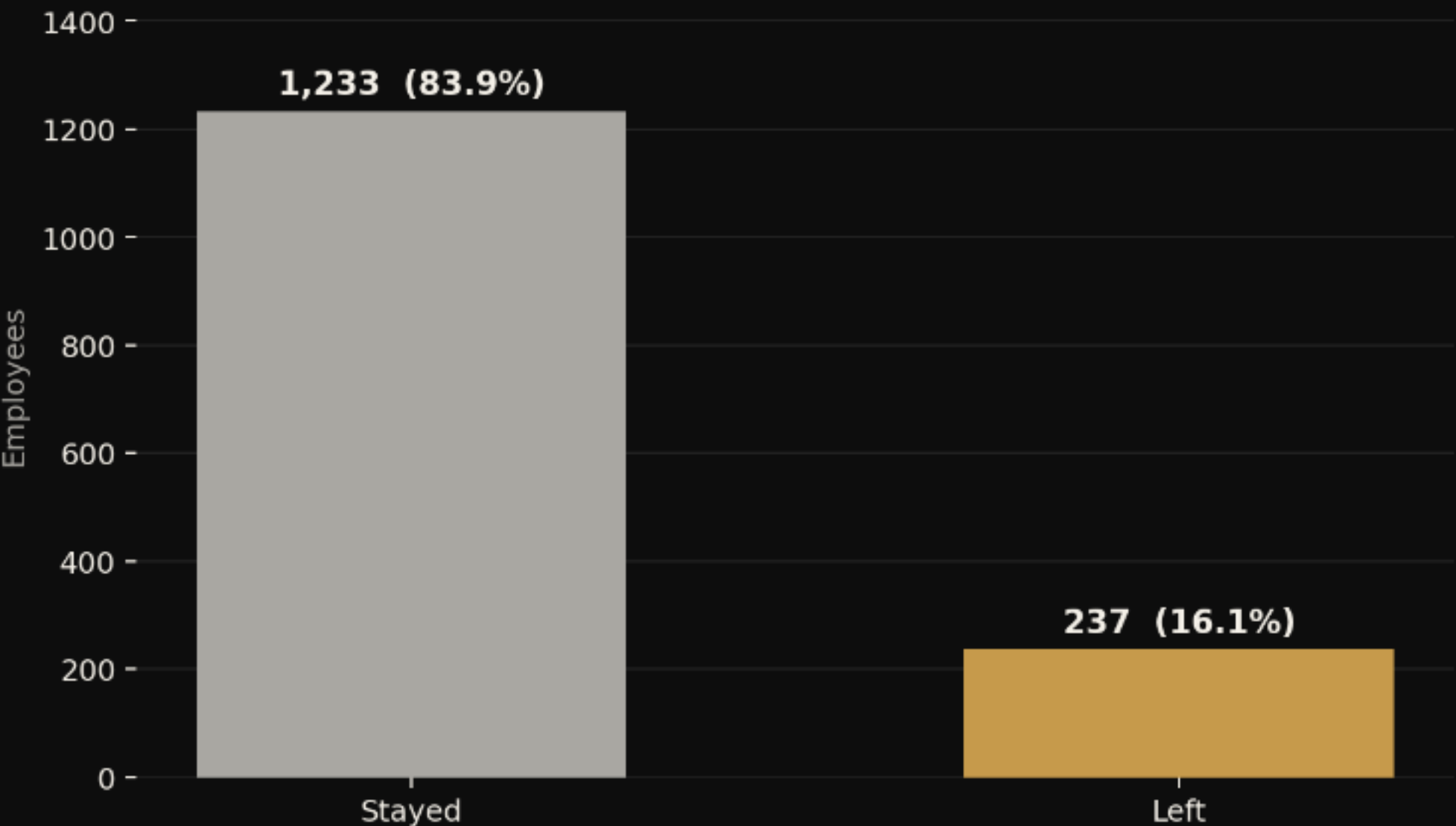
# R&D dominates headcount; three roles anchor staffing



BASELINE

# One in six employees left in this dataset

Overall employee attrition



16.12%

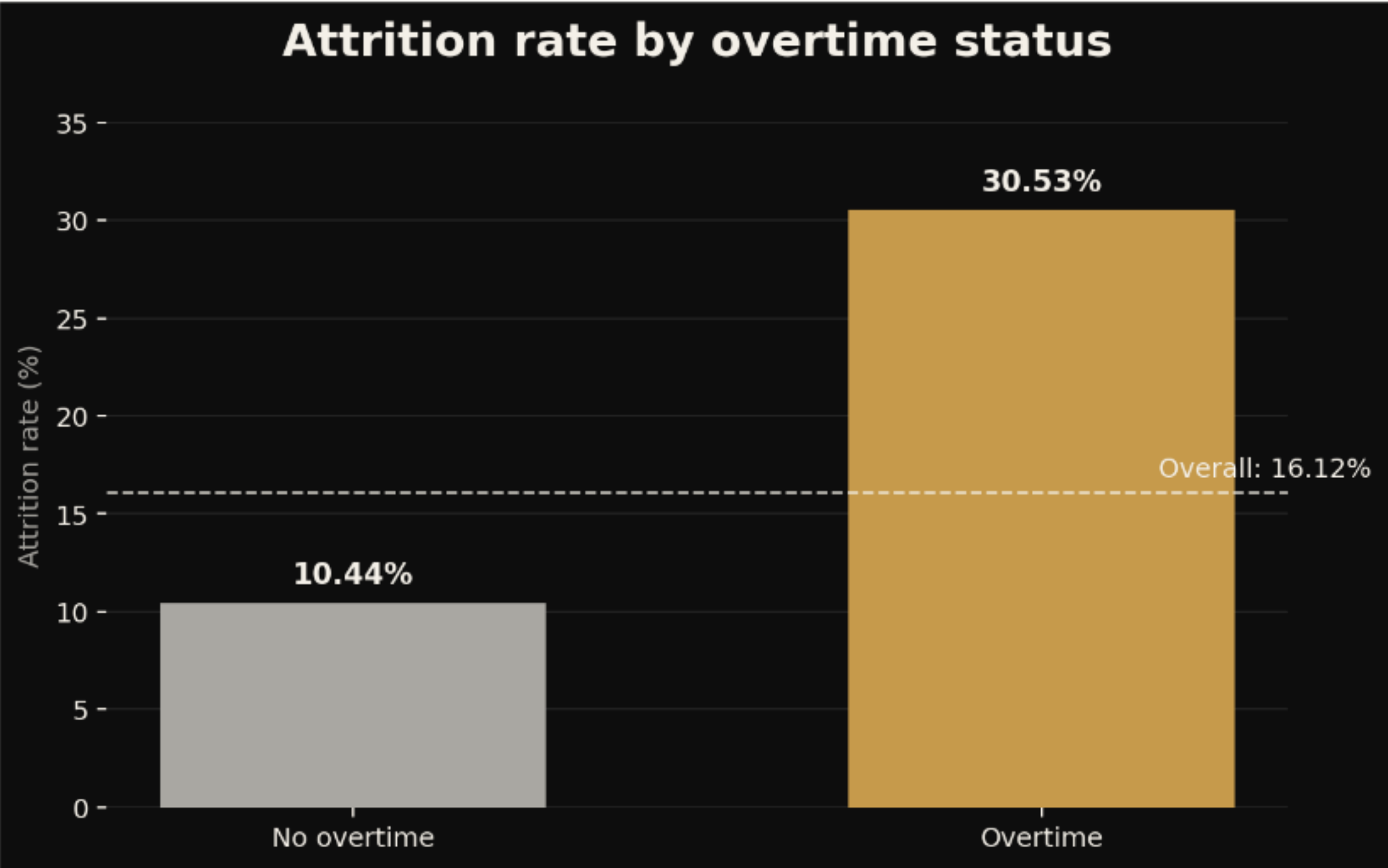
237 of 1,470 employees

This company-wide rate is the reference point for every subgroup. A department, role or condition is considered elevated only when its observed rate sits meaningfully above the 16.12% baseline.



STRONGEST DESCRIPTIVE SIGNAL

# Overtime nearly triples the attrition rate



**30.53%**

Works overtime

**10.44%**

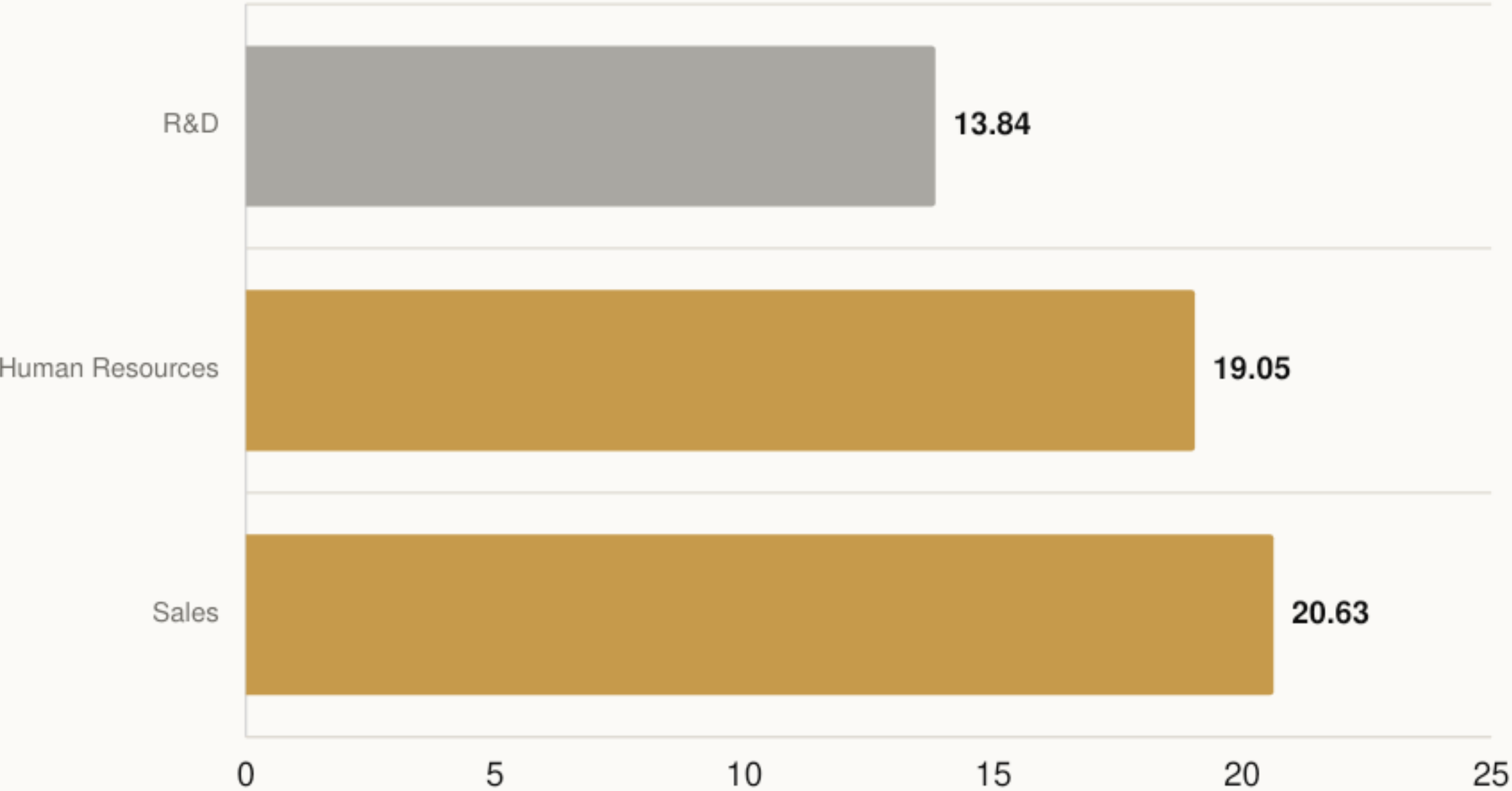
No overtime

**+20.09 percentage points**

Priority for investigation, not proof of causation.

DEPARTMENT CONCENTRATION

# Sales and HR exceed the company baseline



READING THE GAP

**Sales: 20.63%**  
**HR: 19.05%**  
**Overall: 16.12%**  
**R&D: 13.84%**

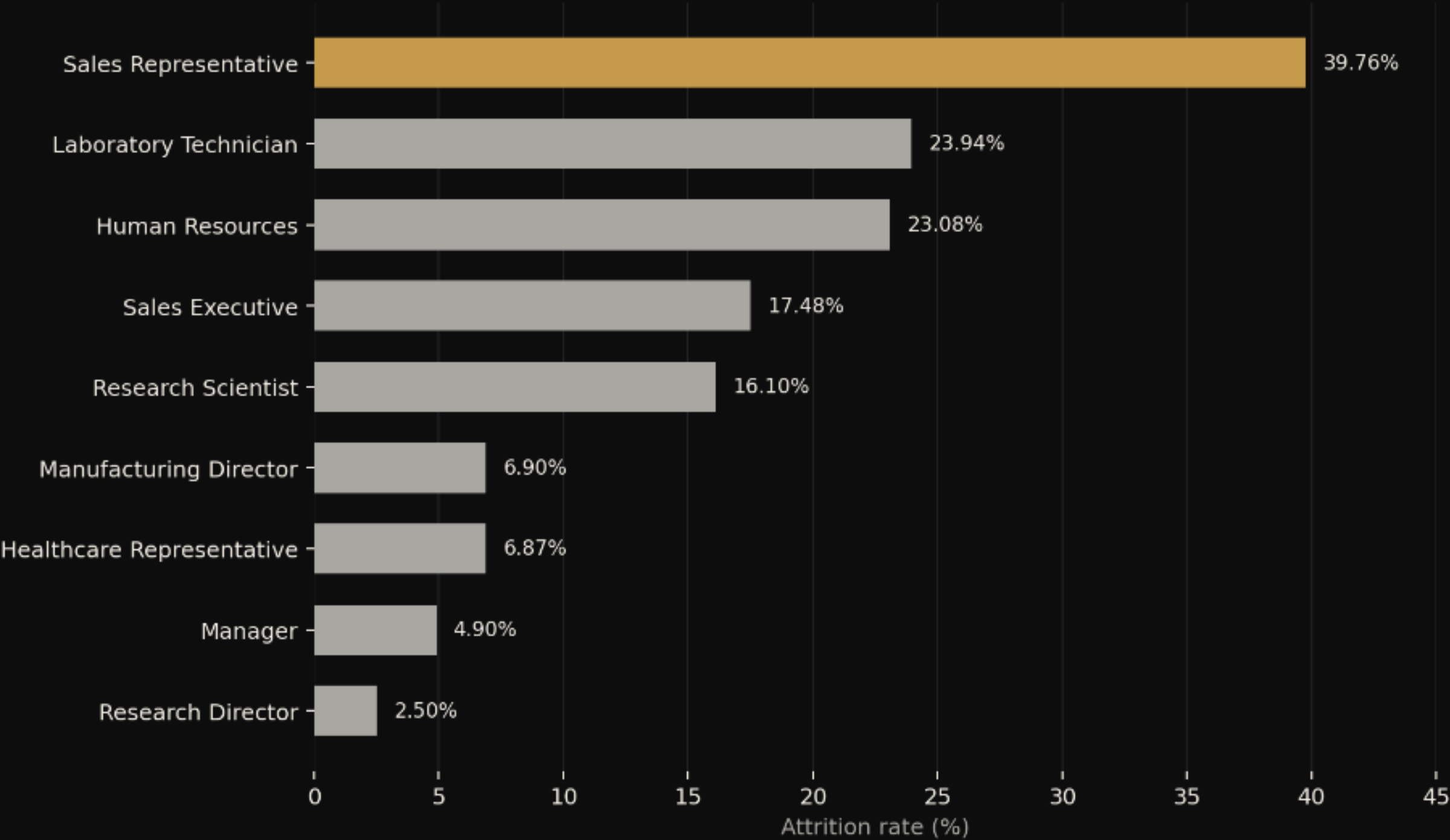
HR contains only 63 employees, so its rate is less stable. Department results should be interpreted with role mix and headcount in view.



SHARPEST DIVIDE

# Attrition concentrates in specific roles

Observed attrition rate by job role



**39.76%**  
Sales Representatives

**23.94%**  
Laboratory Technicians

Role-specific investigation matters more than a single company-wide conclusion.

WHO IS LEAVING?

# Leavers skew younger, lower-paid and less experienced

AGE

33.6

Left company

37.6

Stayed

years

MONTHLY INCOME

4,787

Left company

6,833

Stayed

average

TOTAL WORKING YEARS

8.2

Left company

11.9

Stayed

years

These averages overlap between groups. They are candidate signals to assess jointly with overtime and role, not standalone explanations.



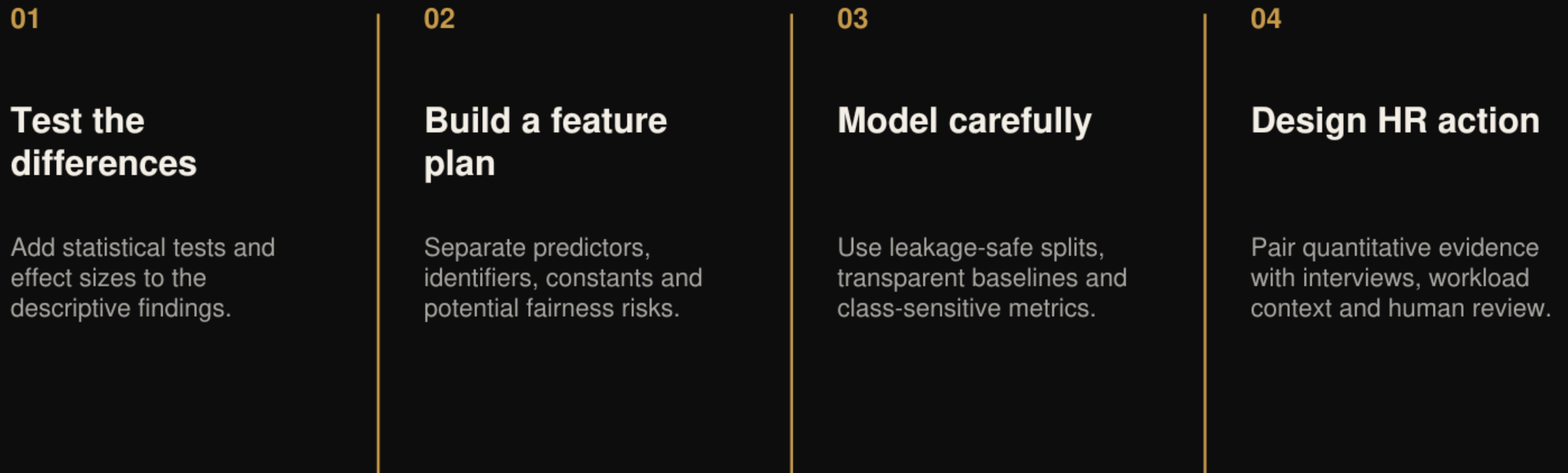
WHAT HR SHOULD INVESTIGATE NEXT

# Five evidence-led priorities from the exploratory analysis

- |    |                           |                                                                        |
|----|---------------------------|------------------------------------------------------------------------|
| 01 | <b>Overtime</b>           | 30.53% attrition versus 10.44% without overtime.                       |
| 02 | <b>Role concentration</b> | Sales Representatives and Laboratory Technicians stand out.            |
| 03 | <b>Department context</b> | Sales and HR exceed the 16.12% company baseline.                       |
| 04 | <b>Career stage</b>       | Leavers are younger and less experienced on average.                   |
| 05 | <b>Compensation</b>       | Lower average income travels with attrition, but does not prove cause. |

CONTINUE FROM EVIDENCE TO RESPONSIBLE MODELLING

# Four steps protect the quality of the next phase



**The strongest descriptive pattern is a starting point for investigation - not proof of causation.**